Chapter 8. Functions Defined by Integrals

Continuity and differentiation under the integral sign, the gamma and beta functions, integral transforms

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8.1 Integrals depending on a parameter

Motivation. Functions like $\Gamma(s) = \int_0^\infty t^{s-1}e^{-t}dt$ are defined by integrals with a parameter. We need conditions under which such functions are continuous and can be differentiated under the integral sign.

For f on $[a,b]\times[c,d]$ let $\phi(x) = \int_c^d f(x,t)\,dt$. Write $D_1 f = \partial f/\partial x$.

Proposition 8.1 (8.1.1)

Let $f : [a,b]\times[c,d] \to \mathbb{R}$ be continuous.
(a) ϕ is continuous on $[a,b]$.
(b) If $D_1 f$ exists and is continuous on $(a,b)\times[c,d]$, then ϕ is differentiable on (a,b) and $\phi'(x) = \int_c^d D_1 f(x,t)\,dt$.

Proof. (a) f is uniformly continuous on the compact rectangle (Theorem 3.18 of Volume I): given ϵ there is δ with $|f(x,t) - f(x',t)| < \epsilon$ for $|x-x'| < \delta$ and all t; then $|\phi(x) - \phi(x')| \le \epsilon(d-c)$.

(b) Fix x and a compact $[x-\eta, x+\eta] \subset (a,b)$, on which $D_1 f$ is uniformly continuous. For $0 < |h| < \eta$, the mean value theorem gives $\frac{f(x+h,t)-f(x,t)}{h} = D_1 f(x+\theta h, t)$ with $0 < \theta = \theta(h,t) < 1$, so

$$\left|\frac{\phi(x+h)-\phi(x)}{h} - \int_c^d D_1 f(x,t)\,dt\right| \le \int_c^d |D_1 f(x+\theta h,t) - D_1 f(x,t)|\,dt \le \epsilon(d-c)$$

once $|h|$ is small. ■

Example 8.1

Differentiating $\int_0^1 t^x\,dt = \frac{1}{x+1}$ n times in $x > -1$ gives $\int_0^1 t^x (\log t)^n dt = (-1)^n n!/(x+1)^{n+1}$. Interchanging the order, $\int_a^b \phi(x)\,dx = \int_c^d \int_a^b f(x,t)\,dx\,dt$ for continuous f: both sides, as functions of the upper limit b, vanish at $b = a$ and have the same derivative $\int_c^d f(b,t)\,dt$ (by (a), (b) and Theorem 5.11 of Volume I).

For f continuous on $[a,b]\times[c,d)$ $(d \le \infty)$, consider the improper integral $\phi(x) = \int_c^d f(x,t)\,dt$.

Theorem 8.2 (8.1.2)

Let $f : [a,b]\times[c,d) \to \mathbb{R}$ be continuous, and suppose there is $g : [c,d) \to \mathbb{R}$ with (a) $|f(x,t)| \le g(t)$ for all (x,t) and (b) $\int_c^d g < \infty$. Then $\int_c^d f(x,t)\,dt$ converges for each x and ϕ is continuous on $[a,b]$.

Proof. Convergence follows from the comparison test (Proposition 7.8). Given ϵ, choose $d' < d$ with $\int_{d'}^d g < \epsilon$. Then $|\phi(x) - \phi_{d'}(x)| < \epsilon$ for all x, where $\phi_{d'}(x) = \int_c^{d'} f(x,t)\,dt$ is continuous by Proposition 8.1. So ϕ is a uniform limit of continuous functions (Theorem 6.1). ■

Theorem 8.3 (8.1.3)

Let $f : [a,b]\times[c,d) \to \mathbb{R}$ be continuous, $D_1 f$ exist and be continuous on $(a,b)\times[c,d)$, and $\int_c^d f(x_0,t)\,dt$ converge for some $x_0 \in (a,b)$. If there is $h : [c,d) \to \mathbb{R}$ with (a) $|D_1 f(x,t)| \le h(t)$ and (b) $\int_c^d h < \infty$, then $\int_c^d f(x,t)\,dt$ converges for each $x \in (a,b)$, and ϕ is differentiable with $\phi'(x) = \int_c^d D_1 f(x,t)\,dt$.

Proof. By the mean value theorem, $|f(x,t)-f(x_0,t)| \le |x-x_0|h(t)$, so $\int_c^d f(x,t)dt$ converges by comparison. For $c < d' < d$ let $\phi_{d'}(x) = \int_c^{d'} f(x,t)dt$; then $\phi'_{d'}(x) = \int_c^{d'} D_1 f(x,t)dt$ (Proposition 8.1), and these converge uniformly in x as $d' \to d$, because the tails are bounded by $\int_{d'}^d h$. Also $\phi_{d'}(x_0) \to \phi(x_0)$. Applying Theorem 6.4 along any sequence $d'_n \to d$ gives the result. ∎

Example 8.2

(a) $F(x) = \int_0^\infty e^{-t^2}\cos(xt)\,dt$ satisfies $F'(x) = -\int_0^\infty te^{-t^2}\sin(xt)\,dt = -\frac{x}{2}F(x)$ (integration by parts), so $F(x) = F(0)e^{-x^2/4} = \frac{\sqrt{\pi}}{2}e^{-x^2/4}$.
(b) $\int_0^\infty \frac{\sin x}{x}dx = \frac{\pi}{2}$: for $G(s) = \int_0^\infty e^{-sx}\frac{\sin x}{x}dx$ $(s>0)$, Theorem 8.3 gives $G'(s) = -\frac{1}{1+s^2}$, and $G(s) \to 0$ as $s \to \infty$, so $G(s) = \frac{\pi}{2} - \arctan s$; continuity at $s=0$ follows from the second mean value theorem (Corollary 5.25 of Volume I).

8.2 The gamma function

Motivation. The gamma function extends the factorial to real arguments and appears throughout probability and analysis. The theorems of the previous section give its basic properties, and the beta integral and Stirling's formula follow.

For $x > 0$ let $\Gamma(x) = \int_0^\infty t^{x-1}e^{-t}\,dt$, convergent since $t^{x-1}e^{-t} \le t^{x-1}$ near 0 and $\le Ct^{-2}$ near ∞. By Theorem 8.3 on compact subintervals of $(0,\infty)$, Γ is C^∞ with $\Gamma^{(n)}(x) = \int_0^\infty t^{x-1}(\log t)^n e^{-t}dt$. Integration by parts gives $\Gamma(x+1) = x\Gamma(x)$, and $\Gamma(1) = 1$, so $\Gamma(n+1) = n!$.

A function $f : I \to \mathbb{R}$ is **convex** if $f((1-\lambda)x+\lambda y) \le (1-\lambda)f(x) + \lambda f(y)$ for $x, y \in I$, $0 \le \lambda \le 1$.

Proposition 8.4 (8.2.1)

(a) A continuous $f : I \to \mathbb{R}$ with $f(\frac{x+y}{2}) \le \frac{f(x)+f(y)}{2}$ for all $x, y \in I$ is convex.
(b) A convex f satisfies, for $s < t < u$ in I,

$$\frac{f(t)-f(s)}{t-s} \le \frac{f(u)-f(s)}{u-s} \le \frac{f(u)-f(t)}{u-t}.$$

Proof. (a) By induction on n, the convexity inequality holds for dyadic $\lambda = k/2^n$: if $\lambda = \frac{\lambda_1+\lambda_2}{2}$ with dyadic λ_i of level $n-1$, then $(1-\lambda)x+\lambda y$ is the midpoint of the points for λ_1, λ_2. For general λ, approximate by dyadic $\lambda_n \to \lambda$ and use continuity.

(b) Write $t = (1-\lambda)s + \lambda u$ with $\lambda = \frac{t-s}{u-s}$; then $f(t) \le (1-\lambda)f(s) + \lambda f(u)$, which rearranges to each of the two inequalities. ∎

$\log\Gamma$ is convex: by Hölder's inequality with exponents $1/\lambda$, $1/(1-\lambda)$ (or Cauchy–Schwarz for the midpoint and Proposition 8.4(a)),

$$\Gamma\Big(\frac{x+y}{2}\Big) = \int_0^\infty t^{\frac{x-1}{2}}e^{-t/2}\,t^{\frac{y-1}{2}}e^{-t/2}\,dt \le \Gamma(x)^{1/2}\Gamma(y)^{1/2}.$$

Theorem 8.5 (8.2.2, Bohr–Mollerup)

If $f : (0,\infty) \to (0,\infty)$ satisfies (a) $f(x+1) = xf(x)$, (b) $f(1) = 1$, (c) $\log f$ is convex, then $f = \Gamma$.

Proof. Γ has these properties, so it suffices to show f is determined by them; by (a) it suffices to determine f on $(0,1]$. Let $\phi = \log f$, $0 < x \le 1$, $n \in \mathbb{N}$. By (a), (b), $\phi(n+1) = \log n!$. Apply

Proposition 8.4(b) to the points $n < n+1 < n+1+x$ and $n+1 < n+1+x < n+2$:

$$\log n \le \frac{\phi(n+1+x)-\phi(n+1)}{x} \le \log(n+1).$$

By (a), $\phi(n+1+x) = \phi(x) + \log[x(x+1)\cdots(x+n)]$. Hence

$$0 \le \phi(x) - \log\frac{n!\,n^x}{x(x+1)\cdots(x+n)} \le x\log\Big(1+\frac{1}{n}\Big) \to 0,$$

so $f(x) = \lim_n \frac{n!\,n^x}{x(x+1)\cdots(x+n)}$ (Gauss's formula) is determined. ∎

Example 8.3

The **beta function** $B(x,y) = \int_0^1 t^{x-1}(1-t)^{y-1}dt$ satisfies $B(x,y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}$: for fixed y, $f(x) = \Gamma(x+y)B(x,y)/\Gamma(y)$ satisfies (a), (b), (c) of Theorem 8.5. In particular $\Gamma(\frac12)^2 = B(\frac12,\frac12) = \pi$, so $\int_{-\infty}^{\infty} e^{-x^2}dx = \Gamma(\frac12) = \sqrt{\pi}$. Stirling's formula $\Gamma(x+1) \sim \sqrt{2\pi x}\,x^x e^{-x}$ follows from the substitution $t = x(1+u)$ and Laplace's method.

8.3 Selected exercises

Motivation. These problems from the textbook's exercise set apply differentiation under the integral sign to evaluate classical integrals, and use the functional equation of Γ and termwise integration to compute with the gamma function.

Exercise 8.4.1 (The Gaussian integral)

Let

$$f(x) = \Big(\int_0^x e^{-t^2}dt\Big)^2, \qquad g(x) = \int_0^1 \frac{e^{-x^2(t^2+1)}}{t^2+1}\,dt.$$

Compute $(f+g)'$, show that $f(x)+g(x) = \frac{\pi}{4}$, and deduce $\lim_{A\to\infty}\int_0^A e^{-t^2}dt = \frac12\sqrt{\pi}$.

Solution.

Step 1. By the fundamental theorem of calculus, $f'(x) = 2e^{-x^2}\int_0^x e^{-t^2}dt$.

Step 2. The integrand $h(x,t) = e^{-x^2(t^2+1)}/(t^2+1)$ is continuous on $\mathbb{R}\times[0,1]$ with $D_1h(x,t) = -2xe^{-x^2(t^2+1)}$ continuous. By Proposition 8.1(b) on every bounded interval,

$$g'(x) = -2e^{-x^2}\int_0^1 xe^{-x^2t^2}dt = -2e^{-x^2}\int_0^x e^{-u^2}du,$$

by the substitution $u = xt$ (valid for every real x, including $x \le 0$).

Step 3. So $(f+g)' = 0$, and $f+g$ is constant, equal to $f(0)+g(0) = 0 + \int_0^1 \frac{dt}{1+t^2} = \frac{\pi}{4}$.

Step 4. Since $0 \le g(x) \le e^{-x^2}\int_0^1 \frac{dt}{1+t^2} = \frac{\pi}{4}e^{-x^2}$, $g(x) \to 0$ and so $f(x) \to \frac{\pi}{4}$ as $x \to \infty$. As $\int_0^x e^{-t^2}dt \ge 0$ for $x \ge 0$, taking square roots gives $\int_0^A e^{-t^2}dt \to \frac{\sqrt{\pi}}{2}$. ∎

Exercise 8.4.4 (A Laplace-type integral)

For $x \in \mathbb{R}$ let

$$f(x) = \int_0^\infty \frac{\cos tx}{1+t^2}\,dt, \qquad g(x) = \int_0^\infty \frac{\sin tx}{t(1+t^2)}\,dt.$$

(a) Show that g is differentiable on $\mathbb{R}$ and $g' = f$.
(b) Show that f is continuous on $\mathbb{R}$.
(c) Show that f is differentiable on $(0,\infty)$ and $g''(x) - g(x) + \frac{\pi}{2} = 0$ for $x > 0$.
(d) Show that $f(x) = \frac{\pi}{2}e^{-|x|}$ for every $x \in \mathbb{R}$.

Solution.

(a) Let $k(x,t) = \frac{\sin tx}{t(1+t^2)}$ for $t > 0$ and $k(x,0) = x$; then k is continuous on $\mathbb{R} \times [0,\infty)$ (as $\frac{\sin tx}{t} \to x$ locally uniformly in x), and $D_1k(x,t) = \frac{\cos tx}{1+t^2}$ is continuous with $|D_1k| \le \frac{1}{1+t^2}$, which is integrable on $[0,\infty)$. Since $|\sin tx| \le |tx|$, $|k(x,t)| \le \frac{|x|}{1+t^2}$, so $g(x)$ converges. Theorem 8.3 on every bounded interval gives $g'(x) = \int_0^\infty \frac{\cos tx}{1+t^2}\,dt = f(x)$.

(b) $\left|\frac{\cos tx}{1+t^2}\right| \le \frac{1}{1+t^2}$, so f is continuous by Theorem 8.2. Also $|f| \le \frac{\pi}{2}$ and $f(0) = \frac{\pi}{2}$.

(c) Fix $x > 0$.

Step 1. Since $\frac{1}{t(1+t^2)} = \frac{1}{t} - \frac{t}{1+t^2}$ and $\int_0^\infty \frac{\sin tx}{t}\,dt = \frac{\pi}{2}$ (substitute $u = tx$ in Example 8.2(b)), the integral $\int_0^\infty \frac{t\sin tx}{1+t^2}dt$ converges as a difference of convergent integrals, and the substitution $u = tx$ gives

$$g(x) = \frac{\pi}{2} - H(x), \qquad H(x) = \int_0^\infty \frac{u\sin u}{x^2+u^2}\,du.$$

Step 2. Integrating by parts, with $\frac{d}{du}\frac{u}{x^2+u^2} = \frac{x^2-u^2}{(x^2+u^2)^2}$ and boundary terms $-\frac{u\cos u}{x^2+u^2}\Big|_0^\infty = 0$,

$$H(x) = \int_0^\infty \frac{(x^2-u^2)\cos u}{(x^2+u^2)^2}\,du.$$

Step 3. The substitution $u = tx$ also gives $f(x) = \int_0^\infty \frac{x\cos u}{x^2+u^2}\,du$. Here $\frac{\partial}{\partial x}\frac{x\cos u}{x^2+u^2} = \frac{(u^2-x^2)\cos u}{(x^2+u^2)^2}$, bounded in absolute value by $\frac{1}{x^2+u^2} \le \frac{1}{a^2+u^2}$ for $x \ge a > 0$, which is integrable on $[0,\infty)$. By Theorem 8.3 on (a,b), for any $0 < a < b$, f is differentiable on $(0,\infty)$ and, by Step 2,

$$f'(x) = \int_0^\infty \frac{(u^2-x^2)\cos u}{(x^2+u^2)^2}\,du = -H(x) = g(x) - \frac{\pi}{2}.$$

Step 4. With $g' = f$ from (a), $g''(x) = f'(x) = g(x) - \frac{\pi}{2}$.

(d)

Step 1. Let $y = g - \frac{\pi}{2}$ on $(0,\infty)$, so $y'' = y$. Put $w = y' + y$; then $w' = y'' + y' = w$, so $(e^{-x}w)' = 0$ and $w = Ce^x$. Then $(e^x y)' = e^x w = Ce^{2x}$, so $y = Ae^x + Be^{-x}$ for constants A, B.

Step 2. $f = y' = Ae^x - Be^{-x}$ is bounded on $(0,\infty)$ by (b), so $A = 0$ and $f(x) = -Be^{-x}$ for $x > 0$. Continuity of f at 0 and $f(0) = \frac{\pi}{2}$ give $-B = \frac{\pi}{2}$.

Step 3. Hence $f(x) = \frac{\pi}{2}e^{-x}$ for $x \ge 0$, and since f is even, $f(x) = \frac{\pi}{2}e^{-|x|}$ for all x. ■

Exercise 8.4.6 (Gamma at half-integers)

Show that $\Gamma\left(n + \frac{1}{2}\right) = \frac{(2n)!}{n!\,4^n}\sqrt{\pi}$ for $n = 0, 1, 2, \ldots$.

Solution. For $n = 0$ this is $\Gamma(\frac{1}{2}) = \sqrt{\pi}$ (Example 8.3). If it holds for n, then $\Gamma(x+1) = x\Gamma(x)$ gives

$$\Gamma\Big(n + \frac{3}{2}\Big) = \frac{2n+1}{2} \cdot \frac{(2n)!}{n!\,4^n}\sqrt{\pi} = \frac{(2n+2)(2n+1)(2n)!}{4(n+1)\,n!\,4^n}\sqrt{\pi} = \frac{(2n+2)!}{(n+1)!\,4^{n+1}}\sqrt{\pi},$$

which is the claim for $n+1$. ■

Exercise 8.4.7 (A series for Γ)

Show that for every $x > 0$,

$$\Gamma(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \cdot \frac{1}{n+x} + \sum_{n=0}^{\infty} \Big(\frac{1}{n!}\int_1^{\infty} t^{-1}e^{-t}(\log t)^n dt\Big)x^n.$$

Solution. Split $\Gamma(x) = \int_0^1 t^{x-1}e^{-t}dt + \int_1^{\infty} t^{x-1}e^{-t}dt$.

Step 1. (The integral over $(0,1]$.) Let $R_N(t) = e^{-t} - \sum_{n\le N} \frac{(-t)^n}{n!}$. On $[0,1]$, $|R_N(t)| \le \sum_{n>N} \frac{1}{n!} =: \rho_N \to 0$. Since $\int_0^1 t^{x-1}dt = \frac{1}{x}$ and $\int_0^1 t^{n+x-1}dt = \frac{1}{n+x}$,

$$\Big|\int_0^1 t^{x-1}e^{-t}dt - \sum_{n\le N} \frac{(-1)^n}{n!(n+x)}\Big| = \Big|\int_0^1 t^{x-1}R_N(t)\,dt\Big| \le \frac{\rho_N}{x} \to 0.$$

Step 2. (The integral over $[1,\infty)$: setup.) For $t \ge 1$, $t^{x-1}e^{-t} = t^{-1}e^{-t}e^{x\log t} = \sum_n \phi_n(t)$ with $\phi_n(t) = \frac{x^n}{n!}t^{-1}e^{-t}(\log t)^n \ge 0$. Each $\int_1^{\infty}\phi_n$ converges, as $\phi_n \le t^{x-1}e^{-t}$.

Step 3. (Upper bound.) For every N, $\sum_{n\le N}\phi_n \le t^{x-1}e^{-t}$, so $\sum_{n\le N}\int_1^{\infty}\phi_n \le \int_1^{\infty} t^{x-1}e^{-t}dt$. The series $\sum_n \int_1^{\infty}\phi_n$ of nonnegative terms therefore converges, with sum at most $\int_1^{\infty} t^{x-1}e^{-t}dt$.

Step 4. (Lower bound.) For $c > 1$, $\log t \le \log c$ on $[1,c]$, so $\sum\phi_n$ converges uniformly on $[1,c]$ (M-test, Corollary 6.3), and by Proposition 6.5

$$\int_1^c t^{x-1}e^{-t}dt = \sum_n \int_1^c \phi_n \le \sum_n \int_1^{\infty}\phi_n.$$

Letting $c \to \infty$ and combining with Step 3, $\int_1^{\infty} t^{x-1}e^{-t}dt = \sum_n \int_1^{\infty}\phi_n$, which is the second series. Adding Step 1 gives the formula. ■